

# SqueakR :: CHEAT SHEET



## Adding new data

1. **Load** an Excel file using:

```
my_file <-  
add_timepoint_data(data_path =  
"path_to_data", t1 = 2, t2 = 12)
```

2. **Score** the loaded dataset using:

```
my_file <-  
score_timepoint_data(data_subset =  
my_file, group = "Drug",  
experimenter = "My Name")
```

3. **Add** the processed dataset to your experiment object:

```
experiment <-  
add_to_experiment(my_file)
```

**Save** the experiment to a specified directory:

```
experiment <-  
save_experiment(experiment =  
experiment, save_path = "save-dir")
```

## Experiment.RData

Your experiment object will be saved to a specified directory using the **save\_experiment()** function.

The file will be saved as an .RData file, and can be loaded into R at any point to continue analysis, or add or remove more data.

Experiment object files can also be loaded into the **SqueakR Dashboard** for visualization.

## SqueakR Pipeline(s)

```
experiment <- semisqueakRpipeline()
```

creates an experiment by looping through a folder of DeepSqueak data, prompting the user for each file's metadata

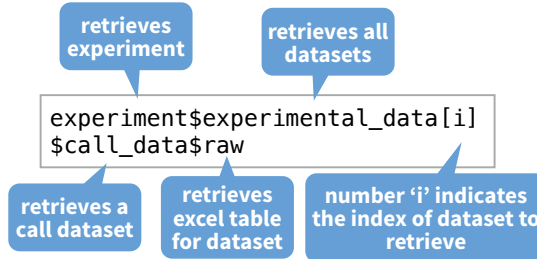
```
experiment <- autosqueakRpipeline()
```

creates an experiment by looping through a folder of DeepSqueak data, pulling metadata from a Google Sheet

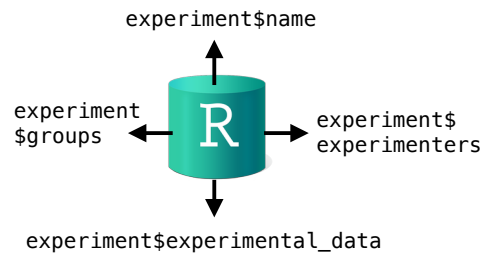
## Managing Experiment

### CODE

How to navigate the experiment object:



### EXPERIMENT OBJECT

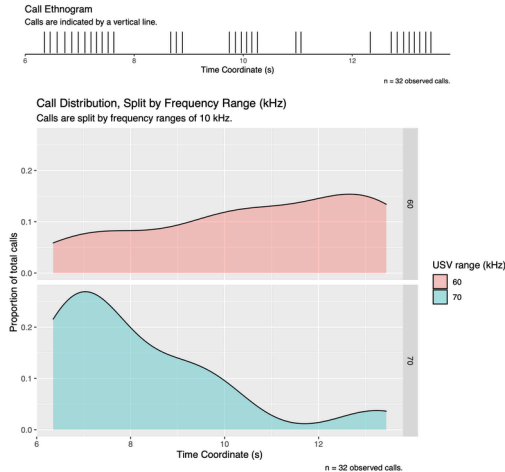


### EXPERIMENT CONTENTS

| item              | description                                        |
|-------------------|----------------------------------------------------|
| name              | the name of the experiment                         |
| last_saved        | the date the object was last saved                 |
| groups            | the experimental groups                            |
| experimenters     | the experimenters who collected data               |
| experimental_data | the full dataset of scored data for the experiment |

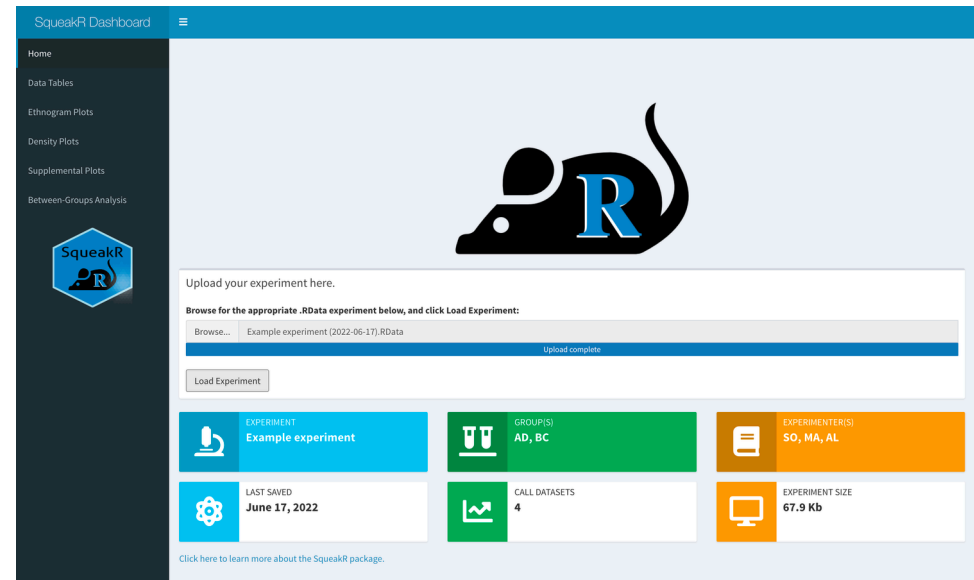
## Visualizations

Graphs which can be generated using SqueakR:



## SqueakR Dashboard

A Shiny dashboard for visualizing and conducting data analysis without any code!



# SqueakR :: CHEAT SHEET



## SqueakR :: Workflow

How to use the SqueakR Dashboard for your experiment:

1. Collect data using a recording software capable of making **.WAV**, **.FLAC**, or **.UVD** audio files.
2. **Detect calls** in the audio files using DeepSqueak (the wiki on how to use the software can be found [here](#)). Analysis can be based on either automated or manual review.
3. **Export the detected call files** as Excel documents to a folder containing all of your experimental data.
4. **Run a SqueakR pipeline** to semi-automatically or fully-automatically generate your experiment **.RData** file.
5. **Load** your experiment into the **SqueakR Dashboard** to visualize and analyze your experiment in a Shiny interface, without any coding required. Data can also be plotted directly into RStudio without the interface, using SqueakR functions.

| Name              | Type                         | Value                |
|-------------------|------------------------------|----------------------|
| experiment        | list [5]                     | List of length 5     |
| name              | character [1]                | 'Example experiment' |
| last_saved        | double (S3: POSIXct, POSIXt) | 2022-06-17 10:45:14  |
| groups            | character [2]                | 'AD' 'BC'            |
| experimenters     | character [3]                | 'SO' 'MA' 'AL'       |
| experimental_data | list [4]                     | List of length 4     |
| call_data         | list [14]                    | List of length 14    |
| call_data         | list [14]                    | List of length 14    |
| call_data         | list [14]                    | List of length 14    |
| call_data         | list [14]                    | List of length 14    |

```
> describe_experiment(experiment)
Experimental name: Example experiment
Last saved: 2022-06-17 10:45:14
Experimenters: SO, MA, AL
Experimental groups: AD, BC
Total call datapoints: 4
Data for AD: 2
Data for BC: 2
>
```

## Workflow Diagram

